

INTERNSHIP PROJECT REPORT

Threaded Subdomain Enumeration Tool using Python

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1. Introduction

Subdomain Enumeration is an important reconnaissance technique used in cybersecurity to identify subdomains associated with a target domain. This project was developed using Python to automate DNS record enumeration and active subdomain discovery using multithreading.

2. Objectives

- Perform DNS record enumeration
- Identify active subdomains
- Automate reconnaissance tasks
- Understand multithreading
- Save discovered subdomains into a file

3. Tools and Technologies Used

- Python
- Requests Library
- DNSPython
- Threading
- Kali Linux

4. Methodology

1. User enters target domain
2. DNS Enumeration performed
3. Wordlist loaded
4. Threaded scanning executed
5. Active subdomains filtered
6. Results saved

5. Results

Example Output:

[✓] Active: <http://www.google.com> (Status: 200)

[✓] Active: <http://mail.google.com> (Status: 200)

[-] Found but not active: <http://api.google.com> (Status: 404)

```
(kali@kali)-[~/Threaded_Subdomain_Enumerator]
$ python subdomain_enum.py
Enter domain: google.com

[+] Starting DNS Enumeration ...

[DNS] A records:
    → 142.250.195.78
[DNS] AAAA records:
    → 2404:6800:4000:1017::66
    → 2404:6800:4000:1017::71
    → 2404:6800:4000:1017::65
    → 2404:6800:4000:1017::8b
[DNS] MX records:
    → 10 smtp.google.com.
[DNS] SOA records:
    → ns1.google.com. dns-admin.google.com. 919691820 900 900 1800 60
```

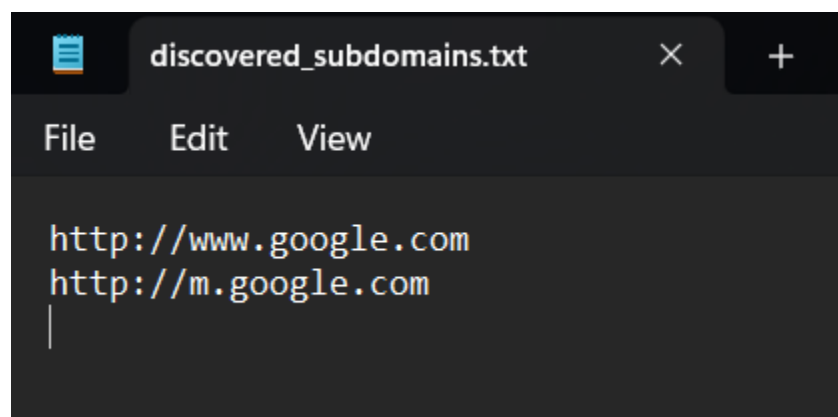
Caption: Figure 1 – DNS Enumeration Output

```
[+] Starting Subdomain Enumeration ...

[-] Found but not active: http://api.google.com (Status: 404)
[✓] Active: http://www.google.com (Status: 200)
[✓] Active: http://m.google.com (Status: 200)

[+] Scan Completed. Results saved in discovered_subdomains.txt
```

Caption: Figure 2 – Active Subdomain Detection



The screenshot shows a text editor window with the title bar 'discovered_subdomains.txt'. The window has a menu bar with 'File', 'Edit', and 'View'. The text area contains the following content:

```
http://www.google.com
http://m.google.com
|
```

Caption: Figure 3 – Saved Results File

6. Applications

- Web Reconnaissance
- Attack Surface Mapping
- Security Research
- Cybersecurity Learning

7. Ethical Considerations

This project is developed strictly for educational and authorized security testing purposes only.

8. Conclusion

The project successfully demonstrates DNS enumeration and threaded subdomain enumeration using Python. Multithreading improved scanning speed while status filtering improved accuracy.